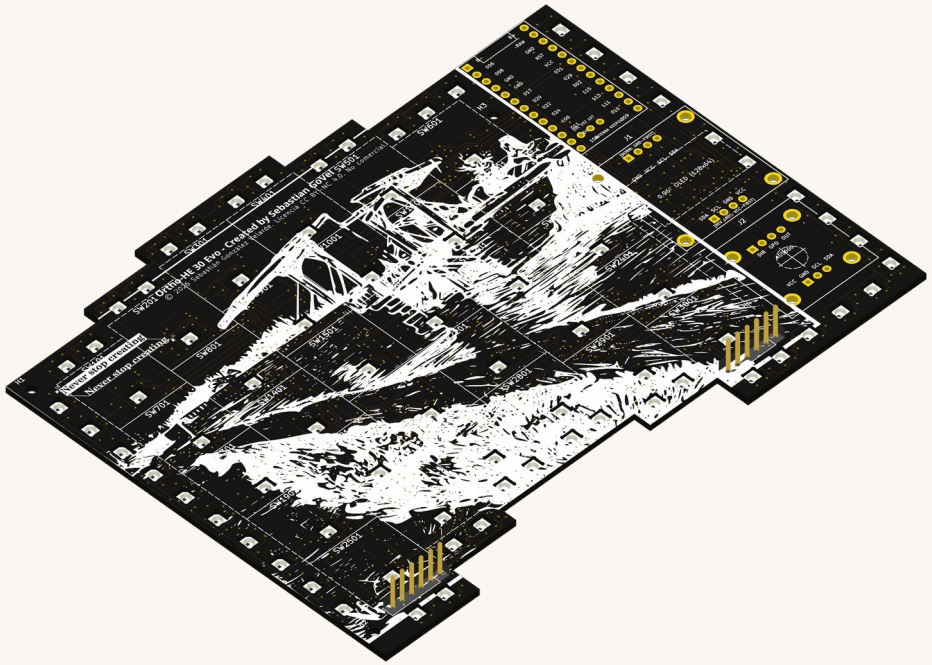
FLOATFORCE

Open-Source Continuous Analog Interface

Redesigning the human-computer interface from first principles:
Springless magnetic switches, native drawbridge tenting, and radical global accessibility.

1517 Fund • Medici Project Pitch | Sebastian Gonzalez V. (Mexico)



MARKET DISRUPTION & OPPORTUNITY

Commercial Keyboards are Overpriced, Locked, & Stagnant

01

Artificial Price Premiums

Big brands charge 3x-5x markups for features that literally cost pennies in silicon. Advanced analog sensors are paywalled behind premium 'gaming' tiers to segment users artificially.

02

Closed Black Boxes

Commercial keyboards are nearly impossible to repair, modify, or inspect. Proprietary software forces bloated background drivers, cloud logins, and zero user autonomy.

03

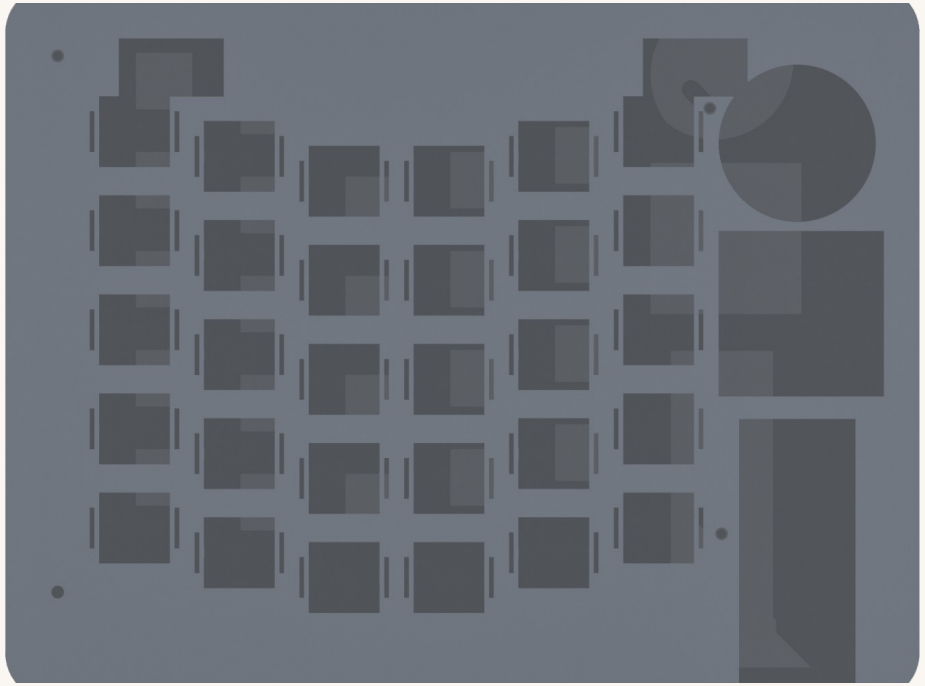
Radical Open Competition

By making advanced magnetic hardware 100% open-source and 3D printable at home, we spur competition, unlock human productivity, and make high-end tools accessible for every language & alphabet.

HUMAN-CENTERED ANATOMY

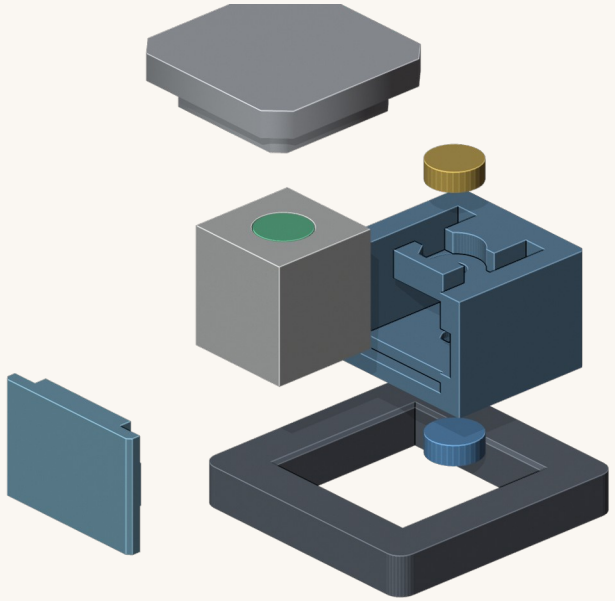
Solving 150 Years of Typewriter Ergonomic Mistakes

- **Horizontally Staggered Keys:**
  A legacy mechanical relic of typewriter levers that forces fingers into unnatural diagonal twists.
- **Hunched Shoulders & Narrow Chest:**
  Traditional unibody keyboards force arms inwards, constricting breathing and posture.
- **Wrist Pronation & Carpal Tunnel:**
  Flat keyboards force hands face-down. Our Drawbridge Tenting (0° to 25°) rotates hands into a natural, neutral handshake posture.
- **Wrist Extension / Painful Angles:**
  Bulky high-profile keyboards bend wrists upwards. Our ultra-low-profile PCB and slim case eliminate wrist extension entirely.



FLOATFORCE V5.0 KINEMATICS

Keys Aren't Binary: 2.36 mm of Continuous Displacement



Springless 3-Neodymium Levitation

Zero metal leaf springs, zero contact chatter, zero mechanical degradation over time. 100% magnetic suspension.

Expressive MIDI Musical Controller

Continuous depth per key enables live synthesizer cutoff sweeps, pitch swells, and vibrato directly on the typing home row.

Proportional 360° WASD Gamepad Joystick

Stealth walk, jog, and full sprint calculated continuously from key depth vectors without external thumbsticks.

3D Navigation & Dual-Stage Triggers

Dynamic orbit controls in CAD/3D software, plus esports Rapid Trigger and dual-action keystrokes (half press = 'a', bottom out = screenshot).

DEMOCRATIZING CREATION

Hardware Engineering from Mexico Powered by AI

AI as an Intellectual Multiplier

• Magnetic Simulation: Parametric modeling in Magpylib to calculate N52 magnetic field saturation and prevent sensor clip.

• Kinematic CAD in Python: 21-step sweep checking static & kinematic collisions (0.0000 mm^3 collision margin).

• KiCad 10 Routing: Dual-bank multiplexed analog bus slashing parasitic line capacitance by 34%.

Accessible Everywhere (No $20k Tooling)

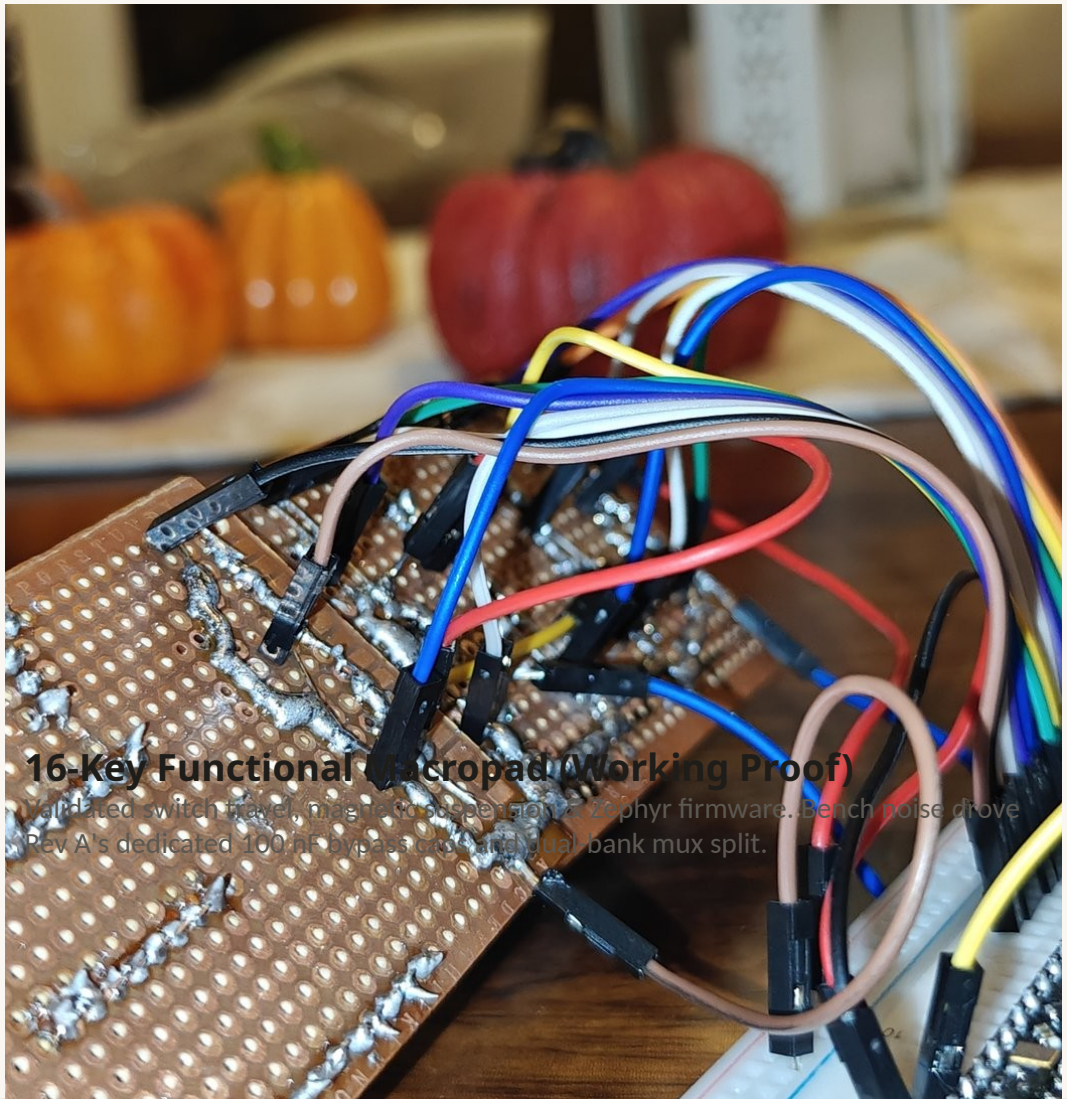
• 100% Support-Free 3D Printing: Keycaps V3.2 angled at -62° and U-Body housing print cleanly on a standard desktop printer without plastic waste.

• Globally Accessible Alphabets: Custom legends, tactile homing, and non-Latin alphabets can be printed on demand.

• Outside Silicon Valley: Proving that ambitious, first-principles hardware can be executed independently from anywhere in the world.

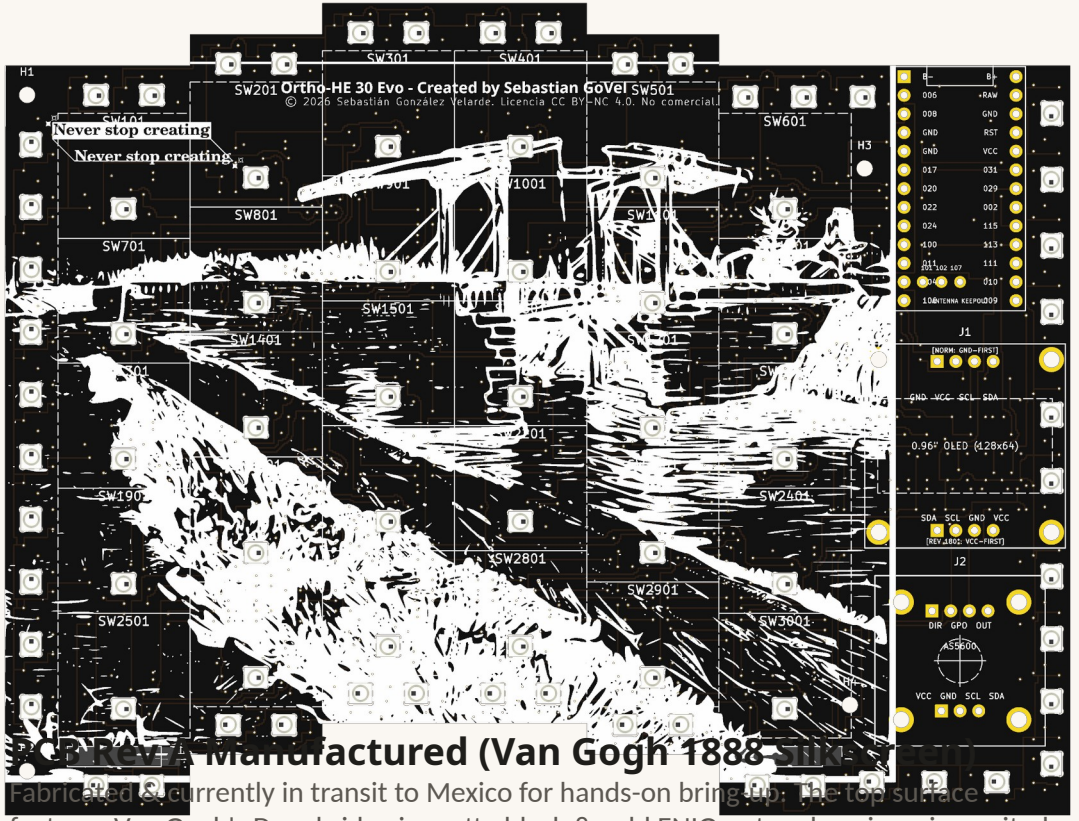
PHYSICAL LABORATORY PROOF

From Bench Prototypes to PCB Rev A: Real Hardware



16-Key Functional Macropad (Working Proof)

Validated switch travel, magnetic suspension & Zephyr firmware. Bench noise drove Rev A's dedicated 100 nF bypass caps and dual-bank mux split.



PCB Rev A Manufactured (Van Gogh 1888 Silkscreen)

Fabricated & currently in transit to Mexico for hands-on bring-up. The top surface features Van Gogh's Drawbridge in matte black & gold ENIG: art and engineering united.

EXECUTION PLAN & SUSTAINABILITY

1517 Fund Medici Grant Allocation & The Prusa Model

Category & Item	Cost	Purpose & Deliverable
1. R&D Runway Recovery	$450.00	Absorb bring-up testing costs & replenish self-funded runway from Rev A.
2. PCB Rev B Iteration	$350.00	Contingent on Rev B design: on-PCB Azoteq trackpad, power gating & parts for 3-5 units.
3. Open Build Guides	$200.00	Filaments for tenting stands, multilingual keycaps, and free beginner video guides.

The 'Prusa Model' in Mexico

- 100% Open Ecosystem: All CAD, firmware (Zephyr), and schematics remain free and forkable under CERN-OHL-P.
- Sustainable Business: Bootstrapping a hardware company selling pre-assembled units, tested PCBAs, and DIY hardware kits directly from Mexico.
- Zero Debt / Zero Bloat: 100% of Medici grant funds go directly to physical hardware R&D.

FUTURE HORIZON

Beyond the Keyboard: The Future of Human-Computer Synergy

VECTOR 01

Frictionless Analog Mice

Continuous pressure-sensitive scrolling and magnetic switches with zero microswitch contact wear.

VECTOR 02

Spatial Bezel-Tracking Glasses

Pupil tracking paired with outward cameras detecting monitor bezels to map natural gaze across screens.

VECTOR 03

Bio-Subtle Interfaces

Detecting ear-canal pressure or micro-facial twitches for hands-free computing and universal accessibility.

This keyboard is only the first chapter. Thank you 1517 team for backing independent, unorthodox builders.